



Name: _____

Teacher: _____

Knox Grammar School 2015

**Trial Higher School Certificate
Examination**

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using blue or black pen only
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11-14, show relevant mathematical reasoning and/or calculations

Teachers:

Mr Vuletich

Mr Bradford

Mrs Dempsey

Ms Yun

Total marks – 70

Section I: Pages 2 – 5

10 marks

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section II: Pages 6 – 10

60 marks

- Attempt Questions 11-14
- Allow about 1 hour 45 minutes for this section

Write your Name, your Board of Studies Student Number and your Teacher's Name on the front cover of each writing booklet

This paper MUST NOT be removed from the examination room

Number of Students in Course: 63

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10.

- 1 The point P divides the interval joining $A(4,1)$ and $B(-1,11)$ externally in the ratio 2:3. Which of these are the coordinates of P ?

- (A) $(2,5)$
- (B) $(14,-19)$
- (C) $(1,7)$
- (D) $(-11,31)$

- 2 What is $\lim_{x \rightarrow 0} \frac{5 \sin 3x}{x}$?

- (A) 15
- (B) $\frac{5}{3}$
- (C) $\frac{3}{5}$
- (D) $\frac{1}{15}$

- 3 Which of the following is the solution to the inequation $\frac{x-3}{x} \leq 0$?

- (A) $x \leq 3$
- (B) $x < 0$ or $x \geq 3$
- (C) $0 < x \leq 3$
- (D) $0 \leq x \leq 3$

4 Differentiate $\frac{1}{\sqrt{4-x^2}}$.

(A) $\sin^{-1} \frac{x}{2}$

(B) $\frac{1}{2} \sin^{-1} \frac{x}{2}$

(C) $\frac{-1}{2\sqrt{(4-x^2)^3}}$

(D) $\frac{x}{\sqrt{(4-x^2)^3}}$

5 The expression $\sin x - \sqrt{3} \cos x$ can be written in the form $2 \sin(x + \alpha)$. Find the value of α .

(A) $\alpha = \frac{\pi}{6}$

(B) $\alpha = -\frac{\pi}{6}$

(C) $\alpha = \frac{\pi}{3}$

(D) $\alpha = -\frac{\pi}{3}$

6 The equation of motion of a particle moving in Simple Harmonic Motion is given by $\ddot{x} = 1 - 3x$. Which of the following statements is true?

(A) The period of motion is $\frac{2\pi}{\sqrt{3}}$ and the centre is $x = \frac{1}{3}$

(B) The period of motion is $\frac{2\pi}{3}$ and the centre is $x = 3$

(C) The period of motion is $\frac{-2\pi}{3}$ and the centre is $x = 3$

(D) The period of motion is $\frac{2\pi}{3}$ and the centre is $x = \frac{1}{3}$.

7 What is the exact value of $\tan^{-1}\left(\tan\frac{5\pi}{6}\right)$?

(A) $-\frac{1}{\sqrt{3}}$

(B) $\sqrt{3}$

(C) $\frac{5\pi}{6}$

(D) $-\frac{\pi}{6}$

8 What is the coefficient of x in $\left(x^2 - \frac{2}{x}\right)^5$?

(A) -160

(B) -80

(C) -32

(D) 3

9 Mark, Greg and four friends arrange themselves at random in a circle. What is the probability that Mark and Greg are not together?

(A) $\frac{1}{120}$

(B) $\frac{2}{5}$

(C) $\frac{3}{5}$

(D) $\frac{119}{120}$

10 By using symmetry arguments, what is the value of $\int_{-a}^a \cos^{-1} x dx$ where $-1 \leq a \leq 1$?

(A) 0

(B) $\frac{a\pi}{2}$

(C) $a\pi$

(D) $2a\pi$

Section II

60 marks

Attempt Questions 11-14

Allow about 1 hour and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11-14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

- (a) Find the size of the acute angle between the lines $2x + y = 5$ and $3x - y = 1$. 2
- (b) Using the substitution $u = 2x^2 + 1$, or otherwise, find $\int x(2x^2 + 1)^{\frac{5}{4}} dx$. 3
- (c) Let $P(x) = (x+1)(x-3)Q(x) + a(x+1) + b$, where $Q(x)$ is a polynomial and a and b are real numbers.
When $P(x)$ is divided by $(x+1)$ the remainder is -11 .
When $P(x)$ is divided by $(x-3)$ the remainder is 1 .
- (i) What is the value of b ? 1
- (ii) What is the remainder when $P(x)$ is divided by $(x+1)(x-3)$? 2
- (d) Let $f(x) = \sin^{-1}(x+4)$.
- (i) State the domain of the function $f(x)$. 1
- (ii) Find the gradient of the graph of $y = f(x)$ at the point where $x = -4$. 2
- (iii) Sketch the graph of $y = f(x)$. 2
- (e) In a factory that makes metal containers it is found that 2% have defects. Find the probability (leaving answers in index form) that a random sample of twenty such metal containers should contain:
- (i) no defective containers. 1
- (ii) not more than one defective container. 1

Question 12 (15 marks) Use a SEPARATE writing booklet.

- (a) Find the exact value of the volume of the solid of revolution formed when the region bounded by the curve $y = \sin 2x$, the x -axis and the line $x = \frac{\pi}{8}$ is rotated about the x -axis. 3
- (b) (i) Find the vertical and horizontal asymptotes of the hyperbola $y = \frac{x-1}{x-3}$ and hence sketch the graph of $y = \frac{x-1}{x-3}$, carefully showing any intercepts with the coordinate axes. 3
- (ii) Hence, or otherwise, find the values of x for which $\frac{x-1}{x-3} \leq 2$ 2
- (c) A team of 4 players consists of 2 men and 2 women.
A total of 7 players are available of which 4 are women and 3 are men.
- (i) How many different teams of 4 players can be selected? 1
- (ii) If 2 of the players are husband and wife and wish to play in the same team, how many different teams can now be selected if the husband and wife are on the team? 1
- (d) Consider the function $f(x) = e^x - e^{-x}$.
- (i) Show that $f(x)$ is increasing for all values of x . 1
- (ii) Show that the inverse function is given by: 3
- $$f^{-1}(x) = \log_e \left(\frac{x + \sqrt{x^2 + 4}}{2} \right)$$
- (iii) Hence, or otherwise, solve $e^x - e^{-x} = 3$. 1
Give your answer correct to two decimal places.

Question 13 (15 marks) Use a SEPARATE writing booklet.

- (a) Use mathematical induction to prove that $3^{3n} + 2^{n+2}$ is divisible by 5, for all positive integers. 3

- (b) (i) Show that the equation $e^{-x} = \sin 2x$ has a root lying between 1 and 2. 1

- (ii) By taking 1.5 as a first approximation, use one application of Newton's method to obtain a better approximation to this root. Give your answer correct to three decimal places. 2

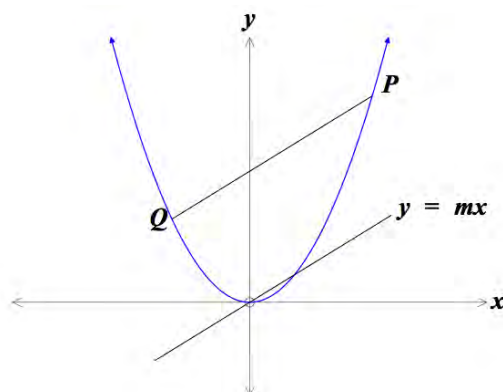
- (c) A particle is moving in simple harmonic motion about the origin O . Its displacement x cm from O at time t seconds satisfies the equation:

$$\ddot{x} = -\pi x$$

- (i) If the velocity of the particle is v cm per second, derive an expression for v^2 as a function of x , given that the amplitude of the motion is 4 cm. 2

- (ii) Where is the particle when its speed is 5 cm per second, giving your answer correct to 4 significant figures? 1

- (d) Two parametric points $P(2p, p^2)$ and $Q(2q, q^2)$ lie on the parabola $x^2 = 4y$, where P lies in the 1st quadrant, Q in the 2nd quadrant and the line through PQ is parallel to the line $y = mx$, where $m > 0$ as shown in the diagram below.



- (i) Show that $p + q = 2m$. 1

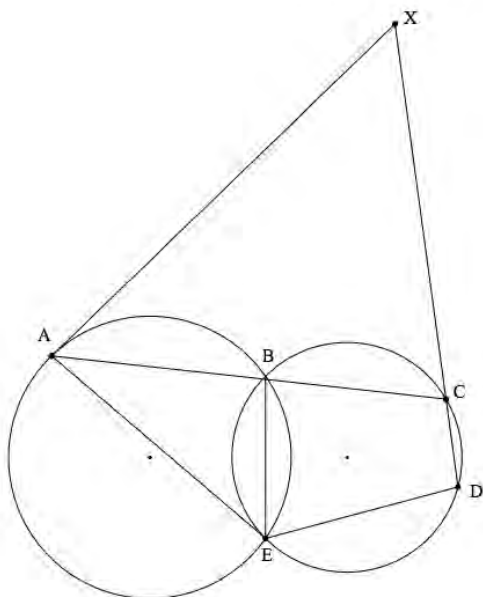
The equation of the normal to the parabola at the point P is $x + py = p^3 + 2p$.
(Do NOT prove this)

- (ii) Find the coordinates of N , the point of intersection of the normals to the curve from P and Q . 2

- (iii) Determine the Cartesian equation of the locus of N as the line PQ moves parallel to the line $y = mx$. Describe the locus of N if $m = 0$. 3

Question 14 (15 marks) Use a SEPARATE writing booklet.

(a)



Two circles intersect at B, E and AX is a tangent to the circle ABE . AB produced meets the second circle at C , and XC meets the circle $BCDE$ again at D as shown.

Let $\angle XAC = \alpha^\circ$ and $\angle AXC = \beta^\circ$.

Copy or trace the diagram into your writing booklet.

- | | | |
|------|--|----------|
| (i) | Give the reason why $\angle AEB = \alpha^\circ$ | 1 |
| (ii) | Prove that $AXDE$ is a cyclic quadrilateral. | 2 |
| | | |
| (b) | Tom and Jerry play a game by each tossing a fair coin. The game consists of tossing the two coins together, until for the first time either two heads appear when Tom wins, or two tails appear when Jerry wins. | |
| (i) | What is the probability that Tom wins at or before the third toss? | 1 |
| (ii) | Hence, or otherwise, show that the probability that Tom wins at or before the n^{th} toss is $\frac{1}{2} - \frac{1}{2^{n+1}}$. | 2 |

Question 14 continues on page 10

Question 14 (continued)

- (c) (i) Use the binomial theorem to write out the expansion of $(1+x)^n$. 1
- (ii) Hence prove that $\sum_{r=1}^n r {}^n C_r 2^{r-1} = n \cdot 3^{n-1}$. 2
- (d) Two particles are projected from the same point on level ground with the same speed V m/s and with angles of projection α and $\frac{\pi}{2} - \alpha$ to the horizontal.

The equations of motion for a particle projected at an angle of α are

$$x = Vt \cos \alpha \text{ and } y = Vt \sin \alpha - \frac{1}{2}gt^2$$

(Do NOT prove these equations)

- (i) Prove that the maximum height h_1 of a particle with angle of projection α 2
is $h_1 = \frac{V^2 \sin^2 \alpha}{2g}$.

The greatest heights they reach are h_1 and h_2 .

- (ii) Show that, for any α , 1
 $h_1 + h_2 = \frac{1}{2}R$, where R is the maximum range.
- (iii) Explain why both particles have the same range. Furthermore, given 3
 $\tan \alpha = \frac{3}{4}$, $V = 200$ m/s and $g = 10$ m/s², what time must elapse between the instants of projection if the particles collide as they strike the ground?

End of paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

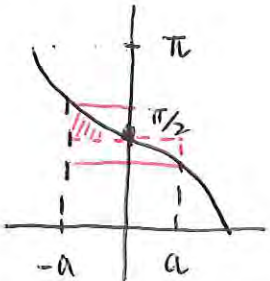
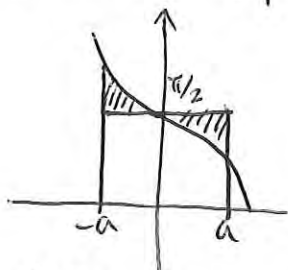
$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

Note $\ln x = \log_e x, \quad x > 0$

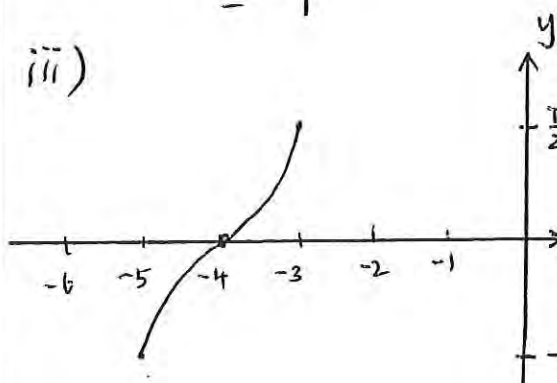
2015 Year 12 Mathematics Ext 1 HSC Task 5 Solutions

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
<u>SECTION I</u> 1) $(4, 1) \quad (-1, 11)$ $\begin{array}{cc} & \times \\ -2 & 3 \end{array}$ $\left(\frac{12+2}{-1}, \frac{-22+3}{1} \right)$ $= (14, -19)$ 2) $5 \times 3 \lim_{x \rightarrow 0} \frac{\sin 3x}{3x}$ $= 15 \times 1$ $= 15$ 3) $\frac{x-3}{x} \leq 0$  $x(x-3) \leq 0$ $0 < x \leq 3 \quad x \neq 0$ 4) $\frac{d}{dx} (4-x^2)^{-\frac{1}{2}}$ $= -\frac{1}{2} (4-x^2)^{-\frac{3}{2}} (-2x)$ $= \frac{x}{\sqrt{(4-x^2)^3}}$ 5) $\sin x - \sqrt{3} \cos x$ $= 2 \sin x \cos x + 2 \cos x \sin x$ $2 \cos x = 1 \quad 2 \sin x = -\sqrt{3}$ $x = -\frac{\pi}{3}$	(B) (A) (C) (D) (D)	6) $\ddot{x} = 1 - 3x$ $= -3(x - \frac{1}{3})$ period $= \frac{2\pi}{\sqrt{3}}$ centre $x = \frac{1}{3}$ 7) $\tan^{-1} [\tan(\frac{5\pi}{6})]$ $= \tan^{-1} (\tan(-\frac{\pi}{6}))$ $= -\frac{\pi}{6}$ 8) $T_{k+1} = {}^5C_k (x^2)^{5-k} (-\frac{2}{x})^k$ $= {}^5C_k x^{10-2k} (-2)^k x^{-k}$ $= {}^5C_k (-2)^k x^{10-3k}$ ie) $10 - 3k = 9$ $k = 3$ $\therefore$ coefficient of x $= {}^5C_3 (-2)^3 = -80$ 9) M, G, A, B, C & D can form a ring of 6 in $5!$ ways seat M - 1 way G - 3 ways seat rest $4!$ ways $P(\text{not together}) = \frac{1 \times 3 \times 4!}{5!}$ $= \frac{3}{5}$	(A) (D) (B) (C)

2015 Year 12 Mathematics Ext 1 HSC Task 5 Solutions

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
10) $\int_{-a}^a \cos^{-1} x \, dx$  because of the symmetry of $\cos^{-1} x$, the area is also equivalent to  area is a rectangle $A = 2a \times \frac{\pi}{2}$ $= a\pi$ SECTION II Question 11 (a) $\tan \theta = \left \frac{m_1 - m_2}{1 + m_1 m_2} \right$ $= \left \frac{(-2) - (3)}{1 + (-2)(3)} \right $ $= 1$ $\therefore \theta = \frac{\pi}{4} \text{ or } 45^\circ$	(C)	(b) $u = 2x^2 + 1$ $\frac{du}{dx} = 4x \text{ ie) } \frac{1}{4} du = x dx$ $\therefore \int x (2x^2 + 1)^{5/4} dx$ $= \int u^{5/4} \cdot \frac{1}{4} du$ $= \frac{1}{4} u^{9/4} \cdot \frac{4}{9} + C$ $= \frac{1}{9} u^{9/4} + C$ $= \frac{1}{9} (2x^2 + 1)^{9/4} + C$ (c) $P(x) = (x+1)(x-3)Q(x) + a(x+1) + b$ i) $P(-1) = -11$ $(-1+1)(-1-3)Q(x) + a(-1+1) + b = -11$ $0 + 0 + b = -11$ $\therefore b = -11$ ii) $P(3) = 1$ ie) $(3+1)(3-3)Q(x) + a(3+1) + b = 1$ $a(3+1) - 11 = 1$ $4a = 12$ $a = 3$ When $P(x)$ is divided by $(x+1)(-3)$, the remainder is $a(x+1) + b$ $3(x+1) - 11$ $= 3x - 8$	Wrong formula -1

2015 Year 12 Mathematics Ext 1 HSC Task 5 Solutions

Suggested Solution (s)	Comments
11(d) $f(x) = \sin^{-1}(x+4)$ i) domain $-1 \leq x+4 \leq 1$ $-5 \leq x \leq -3$ ✓ ii) $f'(x) = \frac{1}{\sqrt{1-(x+4)^2}}$ ✓ $f'(-4) = \frac{1}{\sqrt{1-(-4+4)^2}}$ $= 1$ ✓ iii)  ✓ Range ✓ shape x with x-intercept	
e) $p(0) = \frac{1}{50}$ $p(\tilde{0}) = \frac{49}{50}$ consider $(\frac{49}{50} + \frac{1}{50})^{20}$ i) $P(X=0) = {}^{20}C_0 (\frac{49}{50})^{20}$ $= (\frac{49}{50})^{20}$ ✓ ii) $P(X \leq 1) = P(X=0) + P(X=1)$ $= (\frac{49}{50})^{20} + {}^{20}C_1 (\frac{49}{50})^{19} (\frac{1}{50})$ $= (\frac{49}{50})^{20} + 20 (\frac{49}{50})^{19} (\frac{1}{50})$ $= (\frac{49}{50})^{19} [\frac{49}{50} + \frac{20}{50}]$ ✓ $= (\frac{49}{50})^{19} (\frac{69}{50})$ $(0.98)^{20} + {}^{20}C_1 (0.02)(0.98)^{19}$ or equivalent acceptable.	

QUESTION 12

(a) $V = \pi \int_0^{\frac{\pi}{8}} y^2 dx$
 $= \pi \int_0^{\frac{\pi}{8}} \sin^2 2x dx$ $\cos 2A = 1 - 2\sin^2 A$
 $= \frac{\pi}{2} \int_0^{\frac{\pi}{8}} 1 - \cos 4x dx$ $\sin^2 2x = \frac{1 - \cos 4x}{2}$
 $= \frac{\pi}{2} \left[x - \frac{\sin 4x}{4} \right]_0^{\frac{\pi}{8}}$
 $= \frac{\pi}{2} \left(\frac{\pi}{8} - \frac{1}{4} \sin \frac{\pi}{2} \right) - (0 - \sin 0)$
 $= \frac{\pi}{2} \left(\frac{\pi}{8} - \frac{1}{4} \right)$
 $= \frac{\pi}{16} (\pi - 2) \quad \text{③}$

(b) i) $y = \frac{x-1}{x-3}$ $\frac{\pi^2 - \pi}{16}$

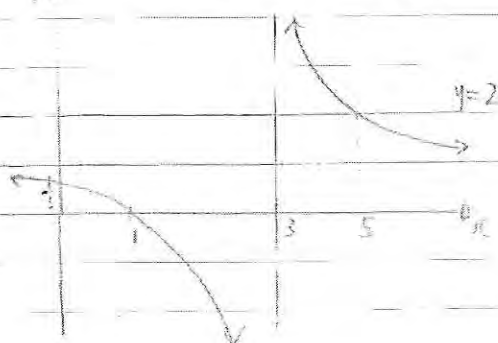
vertical asymptote: $x=3$

$y = \frac{x-3+2}{x-3}$

$= 1 + \frac{2}{x-3}$

as $x \rightarrow \infty$, $\frac{2}{x-3} \rightarrow 0 \therefore y \rightarrow 1$

$\therefore$ horizontal asymptote: $y=1$



At point of intersection with $y=2$

$2 = \frac{x-1}{x-3}$

$2x-6 = x-1$

$x=5$

$\therefore \frac{x-1}{x-3} \leq 2$ for $x < 3$ and $x \geq 5$

(c) 4 players 2 men, 2 women.
 7 available 4M, 3W.

(i) number of teams = ${}^4C_2 \times {}^3C_2$ ①
 $= 18$

(ii) number of teams = ${}^5C_1 \times {}^2C_1$ ①
 $= 6$

(d) $f(x) = e^x - e^{-x}$

(i) $f'(x) = e^x + e^{-x}$
 > 0 for all $x \in \mathbb{R}$. ①
 $\therefore f(x)$ increasing for all $x \in \mathbb{R}$.

(ii) Let $y = e^x - e^{-x}$
 $= e^x - \frac{1}{e^x}$
 $= \frac{e^{2x} - 1}{e^x}$

$f^{-1}: x = \frac{e^{2y} - 1}{e^y}$
 $e^y x = e^{2y} - 1$
 $e^{2y} - e^y x - 1 = 0$

Let $u = e^y$
 $u^2 - ux - 1 = 0$
 $u = \frac{x \pm \sqrt{x^2 - 4(1)(-1)}}{2}$
 $= \frac{x \pm \sqrt{x^2 + 4}}{2}$ ③

$\therefore e^y = \frac{x \pm \sqrt{x^2 + 4}}{2}$

$e^y > 0 \therefore e^y = \frac{x + \sqrt{x^2 + 4}}{2}$

$\therefore y = \ln \left(\frac{x + \sqrt{x^2 + 4}}{2} \right)$

$\therefore f^{-1}(x) = \ln \left(\frac{x + \sqrt{x^2 + 4}}{2} \right)$ as req'd

(iii) $e^x - e^{-x} = 3$
 $f^{-1}(3) = \ln \left(\frac{3 + \sqrt{3^2 + 4}}{2} \right)$ ①
 $= \ln \left(\frac{3 + \sqrt{13}}{2} \right)$
 $= 1.1947$
 $= 1.19 \text{ (2dp)}$

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QUESTION 13.

(a) P.T. $3^{2n} + 2^{n+2}$ is divisible by 5

Step 1: Prove true for $n=1$

$$\text{when } n=1, 3^{2 \cdot 1} + 2^{1+2} = 3^2 + 2^3 = 9 + 8 = 17$$

$$= 17$$

which is divisible by 5

$\therefore$ true for $n=1$

(3)

Step 2:

Assuming true for $n=k$

i.e. $3^{2k} + 2^{k+2} = 5m$ for some integer m

Prove true for $n=k+1$

$$\begin{aligned} 3^{2(k+1)} + 2^{(k+1)+2} &= 3^{2k+2} + 2^{k+3} \\ &= 3^{2k} \cdot 3^2 + 2^k \cdot 2^3 \\ &= 3^{2k} (9) + 2^k (8) \\ &= 3^{2k} (5m - 2^{k+2}) + 2^k \cdot 2^3 \\ &= 135m - 27 \cdot 2^{k+2} + 2 \cdot 2^{k+2} \\ &= 135m - 25 \cdot 2^{k+2} \\ &= 5(27m - 5 \cdot 2^{k+2}) \end{aligned}$$

which is divisible by 5

Since true for $n=1$ and true for $n=k+1$,
assuming true for $n=k$, then true for $n=k+1$,
 $n=2, 3, 4, \dots$ $\therefore$ true for all positive integers

(b) (i) Let $f(x) = e^{-x} - \sin 2x$
 $f(1) = e^{-1} - \sin 2 = -0.541 \dots$
 $f(2) = e^{-2} - \sin 4 = 0.892 \dots$
 $\therefore f(x) < 0$ at $x=1$ and > 0 at $x=2$

$\therefore$ root lies between $x=1$ and $x=2$

(ii) $a_2 = a_1 - \frac{f(a_1)}{f'(a_1)}$ $f'(x) = -e^{-x} - 2\cos 2x$
 $= 1.5 - \frac{e^{-1.5} - \sin 3}{-e^{-1.5} - 2\cos 3}$
 $= 1.453 \dots$ (to 3dp)

(c) $\ddot{x} = -\pi x$

$\ddot{x} = -\pi^2 x$, $\pi \sqrt{\pi}$
 or: $v^2 = \pi^2 (a^2 - x^2)$
 $= \pi (16 - x^2)$

(i) $\frac{1}{2} v^2 = \int -\pi x dx$
 $= -\frac{\pi x^2}{2} + C$

amplitude = 4: when $v=0$, $x=4$

$0 = -\frac{\pi (4)^2}{2} + C$

or when $x=0$

$v^2 = \pi^2 (a^2 - x^2)$

$= \pi^2 (16 - x^2)$

$\therefore v^2 = -\pi x^2 + 16\pi$

(2)

when $x=0$, $v=4\pi$

(ii) when $v=5$, $25 = -\pi x^2 + 16\pi$

$\pi x^2 = 16\pi - 25$

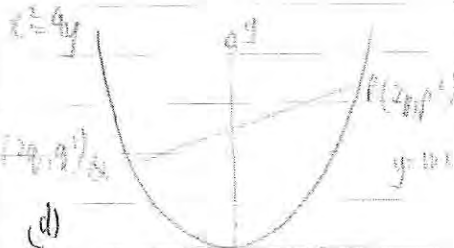
$x^2 = 16 - \frac{25}{\pi}$

$= 16\pi - 25$ (1)

$x = \pm \sqrt{\frac{16\pi - 25}{\pi}}$

$= \pm 2.8358 \dots$

$= \pm 2.836$ (to 3sf)



(d)

$m_{PQ} = \frac{p^2 - q^2}{2p - 2q}$
 $m = \frac{p+q}{2}$

$\therefore p+q = 2m$ as req'd (1)

(ii) $x+py = p^3+2p$ - eqn of normal at P
 At: $x+qy = q^3+2q$ - (2)

$(p-q)y = p^3 - q^3 + 2p - 2q$

$(p-q)y = (p-q)(p^2+pq+q^2) + 2(p-q)$

$y = p^2+pq+q^2+2$

$x + (p^2+pq+q^2+2) = p^3+2p$

$x = -p^2q - pq^2$

$= -pq(p+q)$

(2)

$\therefore N(-pq(p+q), p^2+pq+q^2+2)$

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(iii) for locus of N

$$\begin{aligned}x &= -pq(p+q) \\ &= -pq(2m) \quad \text{from (i)}\end{aligned}$$

$$\begin{aligned}\text{Now, } y &= p^2 + q^2 + pq + 2 \\ &= (p+q)^2 - pq + 2 \\ &= (2m)^2 + \frac{x}{2m} + 2 \\ y &= 4m^2 + \frac{x}{2m} + 2 \quad (2) \\ y &= \frac{x}{2m} + 4m^2 + 2\end{aligned}$$

ψ $m=0$, $m_{pq}=0$ i.e. PQ is horizontal
and $p+q=0$
 $\therefore q=-p$
 $\therefore$ locus of N must be $x=0$.

* Since $p+q=2m$

$$N = (-2mpq, m^2 + 2 - pq)$$

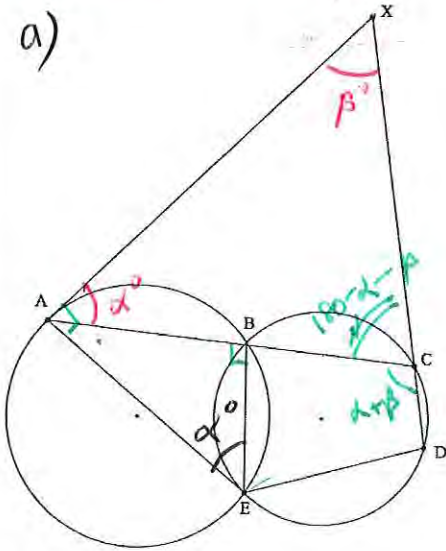
$\therefore$ when $m=0$, $N(0, 2-pq)$

Since $p>0$, $q<0$ (or vice versa), $pq<0$.

$$\therefore 2-pq > 2$$

$\therefore$ locus is $x=0$ for $y>2$.

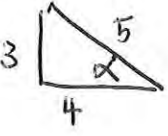
2015 Year 12 Mathematics Ext 1 HSC Task 5 Solutions

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
<u>Question 14</u> a)  i) $\angle AEB = \alpha$ (angle between tangent & chord of contact is equal to the angle in the alternate segment) ii) $\angle XCA = 180 - (\alpha + \beta)$ (angle sum of a triangle) $\angle DEB = 180 - (\alpha + \beta)$ (exterior angle of a cyclic quadrilateral equals interior opposite angle) $\angle AED = \angle AEB + \angle DEB$ $= \alpha + 180 - \alpha - \beta$ $= 180 - \beta$ $\therefore \angle AXC + \angle AED = 180^\circ$ $\therefore AXDE$ is a cyclic quadrilateral as opposite angles are supplementary	✓ ✓	b) i) $\frac{1}{4} + \frac{1}{4}\left(\frac{1}{2}\right) + \frac{1}{4}\left(\frac{1}{2}\right)^2$ $= \frac{7}{16}$ ✓ ii) $\frac{1}{4} + \frac{1}{4}\left(\frac{1}{2}\right) + \frac{1}{4}\left(\frac{1}{2}\right)^2 + \dots + \frac{1}{4}\left(\frac{1}{2}\right)^{n-1}$ It's a G.P. with $a = \frac{1}{4}$ $r = \frac{1}{2}$ $S_n = \frac{a(r^n - 1)}{r - 1}$ ✓ $= \frac{\frac{1}{4}\left[\left(\frac{1}{2}\right)^n - 1\right]}{\frac{1}{2} - 1} = \frac{\frac{1}{4}\left(\frac{1}{2}^n - 1\right)}{-\frac{1}{2}}$ $= \frac{1}{2} \left(1 - \frac{1}{2^n}\right)$ ✓ $= \frac{1}{2} - \frac{1}{2^{n+1}}$ as required. <u>OR</u> Tom & Jerry winning by n^{th} game $P(X) = \frac{1}{2} (1 - P(\text{no one has won}))$ $= \frac{1}{2} \left(1 - \frac{1}{2^n}\right)$ $= \frac{1}{2} - \frac{1}{2^{n+1}}$	

2015 Year 12 Mathematics Ext 1 HSC Task 5 Solutions

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
Question 14 (c) i) $(1+x)^n = \sum_{r=0}^n {}^nC_r x^r$ $= {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \dots + {}^nC_n x^n$ ✓ ii) Differentiating both sides with respect to x $n(1+x)^{n-1}$ $= {}^nC_1 + 2 {}^nC_2 x + 3 {}^nC_3 x^2 + \dots + n {}^nC_n x^{n-1}$ ✓ Now put $x=2$ $n(3)^{n-1} = {}^nC_1 + 2 {}^nC_2 \cdot 2 + 3 {}^nC_3 \cdot 2^2 + \dots + n {}^nC_n 2^{n-1}$ ✓ $n \cdot 3^{n-1} = \sum_{r=1}^n r {}^nC_r 2^{r-1}$		Q14 (d) (ii) greatest height for projection $(\frac{\pi}{2} - \alpha)$ $= \frac{v^2 \sin^2(\frac{\pi}{2} - \alpha)}{2g}$ $= \frac{v^2 \cos^2 \alpha}{2g}$ $h_1 + h_2 = \frac{v^2 \sin^2 \alpha}{2g} + \frac{v^2 \cos^2 \alpha}{2g}$ $= \frac{v^2 (\sin^2 \alpha + \cos^2 \alpha)}{2g}$ ✓ $= \frac{v^2}{2g}$ $= \frac{1}{2} \left(\frac{v^2}{g} \right)$ $= \frac{1}{2} R$ where R is max. range $= \frac{v^2}{g}$.	
Q14 (d) (i) $\dot{x} = v \cos \alpha$ $\dot{y} = v \sin \alpha - gt$ At maximum height $\dot{y} = 0$ ie) $v \sin \alpha - gt = 0$ ✓ $t = \frac{v \sin \alpha}{g}$ — (1) sub (1) into $y(t)$ $y = v \left(\frac{v \sin \alpha}{g} \right) \sin \alpha - \frac{1}{2} g \left(\frac{v \sin \alpha}{g} \right)^2$ ✓ $= \frac{v^2 \sin^2 \alpha}{g} - \frac{v^2 \sin^2 \alpha}{2g}$ $= \frac{v^2 \sin^2 \theta}{2g}$			

2015 Year 12 Mathematics Ext 1 HSC Task 5 Solutions

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
Q14 (d) (iii)  $\cos \alpha = \frac{4}{5}$ $\sin \alpha = \frac{3}{5}$ Time of flight for particle projected @ α $= \frac{2v \sin \alpha}{g}$ $= \frac{2 \times 200 \times \frac{3}{5}}{10}$ $= 24 \text{ sec}$ @ $(\frac{\pi}{2} - \alpha)$ $\Rightarrow \frac{2v \sin(\frac{\pi}{2} - \alpha)}{g}$ $= \frac{2v \cos \alpha}{g}$ $= \frac{2 \times 200 \times \frac{4}{5}}{10}$ $= 32 \text{ seconds}$	✓ ✓	Range for particle with angle of projection α $= \frac{2v^2 \cos \alpha \sin \alpha}{g}$ • angle of $(\frac{\pi}{2} - \alpha)$ $= \frac{2v^2 \sin(\frac{\pi}{2} - \alpha) \cos(\frac{\pi}{2} - \alpha)}{g}$ $= \frac{2v^2 \cos \alpha \sin \alpha}{g}$ = range for particle with angle of projection α ∴ 8 seconds must elapse between the instants of projection if the particle collide so they strike the ground.	✓